

Day 3 — Uttarāyaṇa and Dakṣiṇāyana

Module: Movement of the Sun · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will name the two solstices, identify the half-years they bookend, and use the Sanskrit terms *uttarāyaṇa* and *dakṣiṇāyana* correctly.

Materials

- The calendar from Day 2 (with the four turning-points marked)
- The whiteboard
- A printable two-circle handout (one circle per half-year — see Activity)

Vocabulary introduced today

- *ayana* (IAST: *ayana*; lit. “course, path”) — a half-year course of the Sun, either northward or southward.
- *uttarāyaṇa* (IAST: *uttara-āyaṇa*; lit. “northern course”) — the half-year from the winter solstice (~21 December) to the summer solstice (~21 June). The Sun’s noon altitude *rises* in the Northern Hemisphere over this period.
- *dakṣiṇāyana* (IAST: *dakṣiṇa-āyaṇa*; lit. “southern course”) — the other half-year, ~21 June to ~21 December. The Sun’s noon altitude *falls* in the Northern Hemisphere.
- **Solstice** — Latin *sol-stat*, “Sun stands still.” On the solstice day, the Sun’s noon altitude reaches its highest or lowest value of the year, then begins reversing direction. Sanskrit name: *ayana-samkrānti*.

Lesson flow

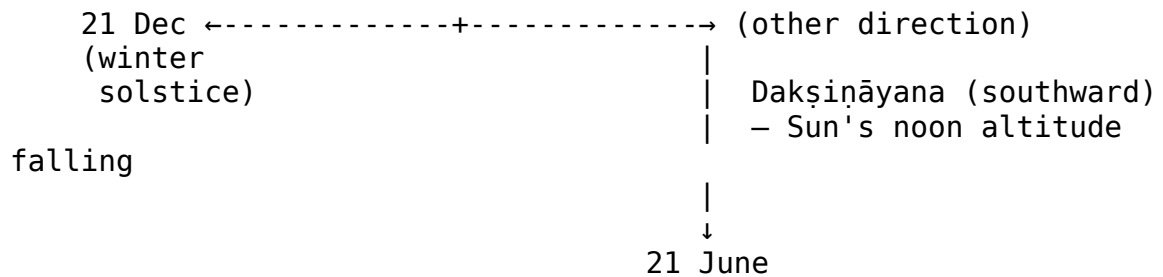
Warm-up (5 min)

- Open the calendar. The four marks from yesterday — March 21, June 21, September 23, December 21 — are visible.
- Prompt: “*Today (insert today’s date) — which ‘turning point’ is closest? And which side of it are we on?*”
- Take 2 responses. They’ll likely give the right answer with help.

Core (20 min)

1. **Two halves of the year.** Sketch on the board:

21 June (summer solstice)
↑
|
| Uttarāyaṇa (northward)
| – Sun’s noon altitude rising
|



No, draw it as a clean cycle:



2. What does “northward” mean exactly?

- The Sun’s **noon position** in the southern sky shifts **NORTH** a little each day during *uttarāyana*.
- In northern India (e.g. Delhi, latitude ~28°N), the noon Sun is **very low in the south** in late December and **almost overhead** in late June.
- “Almost overhead” means the noon shadow of a vertical stick is at its shortest.

3. Paraphrase from the Sūrya-siddhānta (Middle-band citation requirement):

The Indian astronomical text **Sūrya-siddhānta** (compiled in its current form between the 5th and 10th centuries CE) describes the year as having two *ayanas* — *uttarāyana* and *dakṣiṇāyana* — and gives mathematical formulas for the Sun’s daily motion in each. — paraphrased from Sūrya-siddhānta, chapter on solar motion (see [sources.md](#))

4. Tropic of Cancer and Tropic of Capricorn (linking to Geography):

- The latitude at which the Sun is *directly overhead at noon on the June solstice* = 23.5° N = **Tropic of Cancer**. (Cancer = the Latin/Greek name for *Karka rāśi*. The Sun used to be in Karka at the June solstice — Day 7 will explain why “used to be.”)
- The latitude at which the Sun is directly overhead on the December solstice = 23.5° S = **Tropic of Capricorn**. (Capricorn = *Makara*. Same caveat.)
- India straddles the Tropic of Cancer — it runs roughly through Bhopal, Ranchi, and Kolkata.

Activity (15 min) — Calendar marking and formative quiz Part A

Part 1 (8 min) — calendar marking:

- On their marked calendar, students label each half:
 - From Dec 21 → Jun 21: write “**UTTARĀYANA**” in colour
 - From Jun 21 → Dec 21: write “**DAKṢIṆĀYANA**” in colour
- They also write the Sanskrit names next to the four turning-points:
 - Dec 21: *Uttarāyana begins (winter solstice)*
 - Jun 21: *Dakṣiṇāyana begins (summer solstice)*
 - Mar 21 / Sep 23: *Viṣuva* — we’ll meet these tomorrow.

Part 2 (7 min) — Formative Quiz Part A: see quizzes/formative.md. Collect.

Wrap-up (5 min)

- Exit prompt: “*Today (today’s date) — are we in uttarāyana or dakṣiṇāyana? Justify in one sentence.*”
- Preview Day 4: “*Tomorrow — the equinoxes. The days when day and night are equal.*”

Student-facing content

The Sun’s year has two halves:

Half	Sanskrit name	When	What the Sun does (Northern Hemisphere)
North ward	<i>uttarāyana</i>	~21 Dec	Noon altitude rises; days lengthen
		→	
		~21 Jun	
		→	
South ward	<i>dakṣiṇāyana</i>	~21 Jun	Noon altitude falls; days shorten
		→	
		~21 Dec	
		→	

The turning days are the **solstices** (in Sanskrit, *ayana-samkrāntis*).

The latitudes where the Sun is directly overhead at the two solstices have names — **Tropic of Cancer** (23.5° N, on the June solstice) and **Tropic of Capricorn** (23.5° S, on the December solstice). India straddles the Tropic of Cancer.

Homework

- Pick a vertical object near your home that you can see at noon (a flagpole, a tree, a building). For two days this week, at exactly 12:00 noon, look at its shadow. Is the shadow getting *shorter* or *longer*? In a sentence: which *ayana* are we in, and does your shadow observation match?

Differentiation

- **One tier up:** Indian latitude: ~8° N (Kanyakumārī) to ~35° N (Srinagar). For a place at 30° N, how high (in degrees from the horizon) is the noon Sun on the June solstice? On the December solstice? Hint: $90^\circ - \text{latitude} \pm 23.5^\circ$.

- **One tier down:** Focus on the two names. *Uttara* = north. *Dakṣiṇa* = south. *Ayana* = course. That's enough for today.

Teacher notes

- Don't conflate *uttarāyana* with the festival of *Makara Saṁkrānti* yet. Today's *uttarāyana* starts on **December 21** (the modern *sāyana* solstice). *Makara Saṁkrānti* falls on ~**January 14** (the *nirayana* sidereal reckoning). They are off by ~24 days. Why? Day 7 unpacks it. Just hold the puzzle for now if a student raises it.
- The *Sūrya-siddhānta* paraphrase satisfies the Middle-band citation requirement for the module.

Citation(s) used in this lesson

- **Sūrya-siddhānta** (Sanskrit astronomical compendium, current critical text 5th–10th c. CE) — paraphrase of the two-*ayana* division of the solar year. See `sources.md` for the full reference.